Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics

Experience

AI Engineer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web and mobile applications for client businesses via Upwork and Fractal, specializing in full-stack architecture, LLM API integrations, and automated ingestion pipelines.
- **Order Procurement & Management Solution** | *NextJS, NodeJS, PostgreSQL, LLM APIs, REST APIs*
  - Architected a full-stack e-commerce order portal using NextJS, ExpressJS, TypeScript, and PostgreSQL with session-based authentication and 15+ RESTful API endpoints.
  - Designed a dual-tier web portal with a customer ordering UI with real-time inventory listings and an admin suite for role-based task dispatch, audit logging, and recurring order schedules.
  - Built an automated ingestion pipeline using LLM APIs (ChatGPT) to extract structured JSON order data from free-text WhatsApp messages into PostgreSQL, processing 200+ daily orders automatically.
  - Designed relational PostgreSQL schemas with indexed foreign keys and integrity constraints to execute query joins under 50ms across high-volume transaction records.

Software Engineer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Flutter, Python, Django, OpenCV, C* **(Live link)**
  - Built a native C/C++ bridge via `dart:ffi` to port OpenCV SIFT keypoint algorithms into Flutter, enabling single-codebase cross-platform computer vision across Android and iOS devices.
  - Created a real-time camera feed pipeline that normalized OS frame buffers and aspect ratios, converting raw video streams into C structs with zero memory leaks and 30+ FPS performance.
  - Implemented real-time keypoint extraction and feature matching, transmitting match coordinates asynchronously to Dart for dynamic augmented reality UI overlay rendering.
  - Developed mobile user interfaces in Flutter from Figma designs and built Python RESTful APIs in Django to power location-aware spatial discovery features for alpha release.
- **Kia National Conference App** | *Flutter, REST APIs*
  - Engineered an event management mobile app in Flutter for 500+ attendees at the Kia National Dealer Conference in Bangkok, featuring real-time registration, attendee directories, and venue navigation.

Technical Skills

AI & Languages: Python, TypeScript, JavaScript, SQL (PostgreSQL), NodeJS, C++, Dart, C, HTML/CSS
Frameworks & APIs: NextJS, React, ExpressJS, Django, Flutter, NestJS, React Native, RESTful APIs
AI & Data Engineering: Retrieval-Augmented Generation (RAG), LLM APIs, Vector Databases, OpenCV, Pandas, NumPy
Tools & Deployment: AWS, Docker, Git, Linux, Bash, CI/CD, Cursor, Copilot, LaTeX, Agentic Workflows

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)